```
def visualize_gp_belief_and_policy(
    model, likelihood, policy=None, next_x=None
):
    with torch.no_grad():
        predictive_distribution = likelihood(model(xs))   ❶
        predictive_mean = predictive_distribution.mean   ❶
        predictive_upper, predictive_lower =   ❶
          <linearrow /> predictive_distribution.confidence_region()   ❶

        if policy is not None:   ❷
            acquisition_score = policy(xs.unsqueeze(1))   ❷

    ...   ❸
```

❶ GP predictions.

❷ Acquisition score.

❸ Currently omitted; see below.

Here, we are generating the predictions from the GP and the acquisition scores on the test data `xs`. Note that we don't compute the acquisition scores if `policy` is not passed in, in which case we also visualize the GP predictions in the way we're already familiar with: scattered points to indicate the training data, a solid line for the mean predictions, and shaded regions for the 95% credible intervals.

```
   if policy is None:
       plt.figure(figsize=(8, 3))

       plt.plot(xs, ys, label="objective", c="r")   ❶
       plt.scatter(train_x, train_y, marker="x", c="k",
           label="observations")   ❷

       plt.plot(xs, predictive_mean, label="mean")   ❸
       plt.fill_between(   ❸
           xs.flatten(),   ❸
           predictive_upper,   ❸
           predictive_lower,   ❸
           alpha=0.3,   ❸
           label="95% CI",   ❸
       )   ❸

       plt.legend()
       plt.show()
```

❶ Ground truth.

❷ Training data.

❸ Mean predictions and 95% credible intervals.

SIDEBAR **SIDEBAR**

Refer to Section 2.4.4 for a refresher on this visualization.

If, on the other hand, a policy object is passed in, we will create another subplot to show the acquisition scores across the search space:

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>